



The volcanic history of Mars: High-resolution crater-based studies of the calderas of 20 volcanoes

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ARTICLE INFO

Article history:

Received 11 May 2010

Revised 1 November 2010

Accepted 9 November 2010

Available online 19 November 2010

Keywords:

Mars
Cratering
Mars, Surface
Volcanism

ABSTRACT

Determining absolute surface ages for bodies in the Solar System is, at present, only possible for Earth and Moon with radiometric dating for both bodies and biologic proxies such as fossils for Earth. Relative ages through cratering statistics are recognized as one of the most reliable proxies for relative ages, calibrated by lunar geologic mapping and Apollo program sample returns. In this work, we have utilized the Mars Reconnaissance Orbiter's ConTeXT Camera's images which provide the highest resolution wide-scale coverage of Mars to systematically crater-age-date the calderas of 20 of Mars' largest volcanoes in order to constrain the length of time over which these volcanoes – and major volcanic activity on the planet, by extension – were active. This constitutes the largest uniform and comprehensive research on these features to date, eliminating unknown uncertainties by multiple researchers analyzing different volcanoes with varied data and methods. We confirm previous results that Mars has had active volcanism throughout most of its history although it varied spatially and temporally, with the latest large-scale caldera activity ending approximately 150 ma in the Tharsis region. We find a transition from explosive to effusive eruption style occurring in the Hesperian, at approximately 3.5 Ga ago, though different regions of the planet transitioned at different times. Since we were statistically complete in our crater counts to sizes as small as ~60 m in most cases, we also used our results to study the importance of secondary cratering and its effects on crater size–frequency distributions within the small regions of volcanic calderas. We found that there is no “golden rule” for the diameters secondaries become important in crater counts of martian surfaces, with one volcano showing a classic field of secondaries ~2 crater diameters from the center of its primary but not affecting the size–frequency distribution, and another clearly showing an influence but from no obvious primary.

Published by Elsevier Inc.

1. Introduction

The planet Mars has had an active geologic past with a varied surface showing evidence of past fluvial, aeolian, and volcanic activity. This is in stark contrast with the current lack of geologic activity on Mars, though recent data have illustrated a few lingering aeolian and cryogenic processes that continue to significantly shape its surface (e.g., Hartmann, 2005). Noticeably missing from the list of current activity is volcanism, and an ongoing question has been when the last volcanic episodes occurred on Mars.

Many researchers in the past have tried to answer when volcanism ended through two methods. The method that we will not be addressing in this paper involves radiometric dating of the known martian meteorites, which generally date from a few hundred Ma

to about 1.5 Ga, the noticeable exception being ALH84001 which dates to ~4.5 Ga (Nyquist et al., 2001) though recent estimates place it at ~4.1 Ga (Lapen et al., 2010). The basaltic meteorites (ALH84001 is an orthopyroxenite) require volcanism at the time of formation, constraining the most recent episode on Mars to a maximum of 180 myr ago (the basaltic shergottite meteorites Shergotty, Zagami, and Los Angeles) (Bouvier et al., 2005).

The second method of age dating the martian volcanoes is crater-based, an established technique that assumes an older surface will have a higher crater density because it has had more time to accumulate craters (Arvidson et al., 1979). Since the original images from the Mariner 9 and Viking orbiters were returned in the late 1960s and throughout the 1970s, different researchers have tried to use crater densities to successively refine the ages of the major volcanoes on the planet (e.g., Blasius, 1976; Plescia and Saunders, 1979; Neukum et al., 1979; Neukum and Hiller, 1981; Greeley and Spudis, 1981; Plescia, 1994; Hodges and Moore, 1994 and references therein). In general, the utility of crater counting is greatly increased when (a) larger areas are used and (b) smaller craters are included for the joint effect of better number

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statistics and a larger range of diameters for age determinations. The latter is limited usually by the data resolution for airless or nearly-airless bodies, such as Mars or the Moon.

The last 15 years has brought more orbiters to Mars with ever-higher-resolution cameras, allowing crater-based age dating of the volcanoes to be refined by dating individual features such as flows, collapses, and calderas (e.g., Hartmann, 2005). It has only been in the last ~6 years that the planetary community has had sufficient imagery to derive statistically meaningful ages from areas that are as small as volcanic calderas on the planet, mainly through the use of Mars Express' High Resolution Stereo Camera (HRSC, roughly 10–30 m/pix resolution) and Mars Reconnaissance Orbiter's (MRO's) ConTeXT Camera (CTX, 5.5–7.5 m/pix resolution), resulting in works such as Neukum et al. (2004), Werner (2009), and this work. Previous work with Mars Global Surveyor's Mars Orbiter Camera (MOC) by different researchers in localized parts of the planet that illustrates some lava flows are as young as a few 10s of millions of years (e.g., Berman and Hartmann, 2002; Hartmann, 2005; Roberts et al., 2007).

Neukum et al. (2004) used HRSC images to age-date five volcanic summits – Hecates and Albor Tholi located in Elysium, and Arsia, Ascraeus, and Olympus Montes located in the vast Tharsis region (Fig. 1). Their work is comparable to ours except in number of volcanoes dated and our completeness to decameter-scale craters. Since then, the only researcher to attempt to systematically crater-age-date the volcanic calderas was Werner (2009) who dated all 24 major volcanic constructs using HRSC data, but often all calderas were combined into one, or the calderas were not dated in favor of analyzing the flanks or other flows from the volcanoes.

Our aim was to use the CTX dataset, which offers up to 6× higher resolution than HRSC, to systematically crater-age-date recognizable calderas in the 24 major volcanoes on Mars (Fig. 1). The previous work to systematically date the volcanoes (Werner, 2009) used HRSC, while all previous research was done by different people using various datasets, lacking much consistency when looking at absolute ages at the 10–100 myr precision level. This limits their utility in providing a timeline for volcanism on the planet. In this work, we were able to date 93 calderas within 20 volcanoes using a uniform, high-resolution dataset, and all the crater counting was done by a single person; the four volcanoes

southeast of the Hellas impact basin (Amphitrites, Malea, Peneus, and Pityusa Paterae) have inadequate CTX coverage for this approach as of May 2010 and were not analyzed. We use these data to (a) confirm previous results of continuous volcanism throughout Mars' history until the recent past, (b) argue for dwindling and increasingly localized volcanism through time, (c) discuss the implications of the ages for the eruptive history of the Tharsis region, (d) provide a transition time from explosive to effusive volcanism, and (e) explore ancillary issues of secondary cratering and resurfacing rates.

We discuss our methods of mapping the main volcanic calderas on Mars and crater-age dating in Section 2. Sections 3–7 are devoted to our analysis and results for each volcano with each section describing a different province: Tharsis (Section 3), Syrtis Major (Section 4) and Elysium (Section 5) regions, Apollinaris Mons (Section 6), and the volcanoes surrounding Hellas Basin (Section 7). We then use these results to discuss implications for secondary cratering and resurfacing in Section 8, and finally the implications for the most recent volcanism in various regions of the planet in Section 9. We list the areas of calderas and number of craters identified in Table 1, and we summarize our age results in Table 2.

2. Method

2.1. Image processing

We obtained the highest resolution visible images available with the widest coverage from a single source, the CTX dataset (Malin et al., 2007) from the MRO spacecraft. This instrument records greyscale images of the martian surface at resolutions in the range ~5.5–7.5 meters per pixel (m/pix) that are publicly available through NASA's online Planetary Data Systems (PDS) website (<http://www.pds-imaging.jpl.nasa.gov/>). We obtained the best images in terms of data quality and with the most complete coverage available for our study areas as of May 2010.

The images were processed using the United States Geological Survey's (USGS's) Integrated Software for Imagers and Spectrometers (ISIS) v 3.1 software. They were mosaicked together, projected to a standard equicylindrical spheroid, and output at a final, uniform resolution that was slightly down-sampled from the original to 10 m/pix. This was done to limit the final file sizes to a few

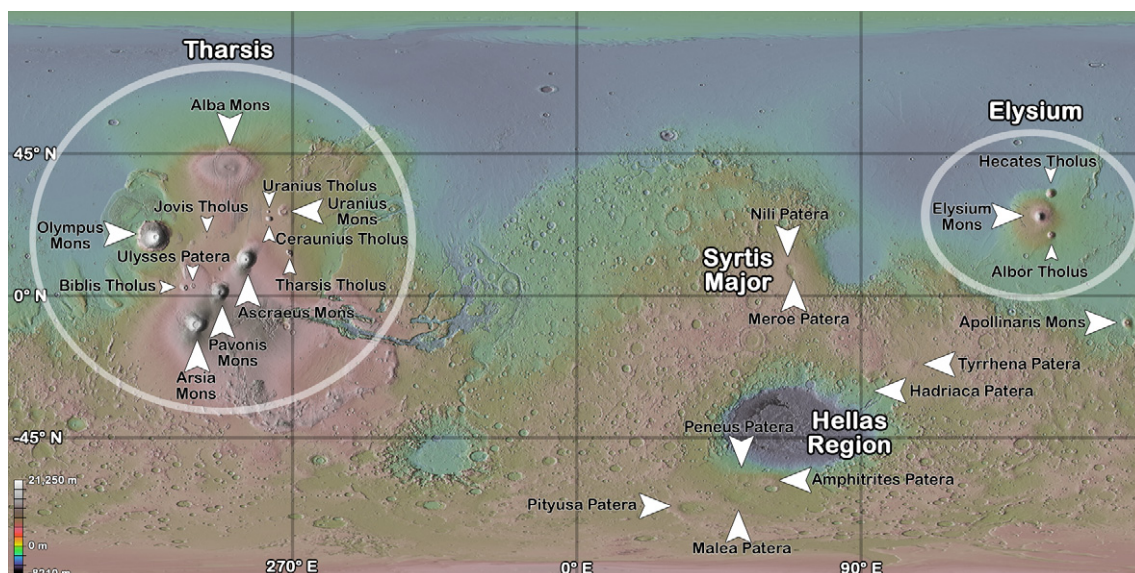


Fig. 1. Underlying map is MOLA (Mars Orbiter Laser Altimeter) shaded relief (Smith et al., 2001), with the arrows and labels showing the locations of the two dozen major identified volcanoes on the planet. Arrow orientation and size is not coded to anything. All volcanoes except those southeast of Hellas Basin were analyzed.